
Lesson 5
BIOMES OF THE EARTH

LEARNING OBJECTIVES

By the end of this lesson, students should be able to:

1. Explain what a biome is and describe its key characteristics.
2. Explain how climate, soil, sunlight, and water influence the distribution and characteristics of biomes.
3. Discuss how human activities affect both terrestrial and aquatic ecosystems.

BIOMES

Biomes: Large ecological areas with **distinct climate, flora, and fauna**.

Two Main Types:	
Terrestrial Biomes (land-based)	Aquatic Biomes (water-based)

TERRESTRIAL BIOMES



TUNDRA

- Cold
- Treeless
- Permafrost soil



TAIGA/BOREAL FOREST

- Coniferous forests
- Long winters

Coniferous trees: cone-bearing trees with needle-like leaves, staying green throughout the year.



TEMPERATE FOREST

- Four seasons
- Deciduous trees

Deciduous trees: Trees with broad leaves that change color and fall off in autumn, regrowing in spring.



GRASSLAND

- Dominated by grasses
- Few trees.



DESERT

- Very dry
- Extreme temperatures.



TROPICAL RAINFOREST

- Warm
- Wet
- High biodiversity

TERRESTRIAL BIOMES: FLORA & FAUNA

Terrestrial Biome	Flora (Plants)	Fauna (Animals)
Tundra	Mosses, lichens, grasses	Arctic fox, caribou, polar bear, snowy owl
Taiga/Boreal Forest	Conifers (pine, spruce, fir), mosses	Moose, lynx, bears, wolves, squirrels
Temperate Forest	Deciduous trees (oak, maple, beech), shrubs, ferns	Deer, foxes, raccoons, black bears, songbirds
Grassland	Grasses, wildflowers, few trees	Bison, antelope, zebras, lions, prairie dogs
Desert	Cacti, succulents, drought-resistant shrubs	Camels, lizards, snakes, scorpions, fennec fox
Tropical Rainforest	Orchids, ferns, broadleaf trees, vines	Jaguars, toucans, monkeys, frogs, insects

- AQUATIC BIOMES

  Freshwater

  Marine

FRESHWATER

Low salinity, supporting diverse plants and animals adapted to non-salty environments.

1. **Rivers**: **Flowing** bodies of freshwater.
2. **Lakes:** Large, **still** freshwater basins.
3. **Wetlands: Water-saturated** ecosystems with high biodiversity.

FRESHWATER : FLORA & FAUNA

Ecosystem	Flora (Plants)	Fauna (Animals)
Rivers	Algae, water grasses, willow trees along banks	Trout, salmon, otters, frogs, kingfishers
Lakes	Water lilies, cattails, reeds, phytoplankton	Ducks, turtles, fish (bass, perch), snails
Wetlands	Mangroves, sedges, bulrushes, mosses	Frogs, herons, dragonflies, beavers, alligators



Rivers



Lakes



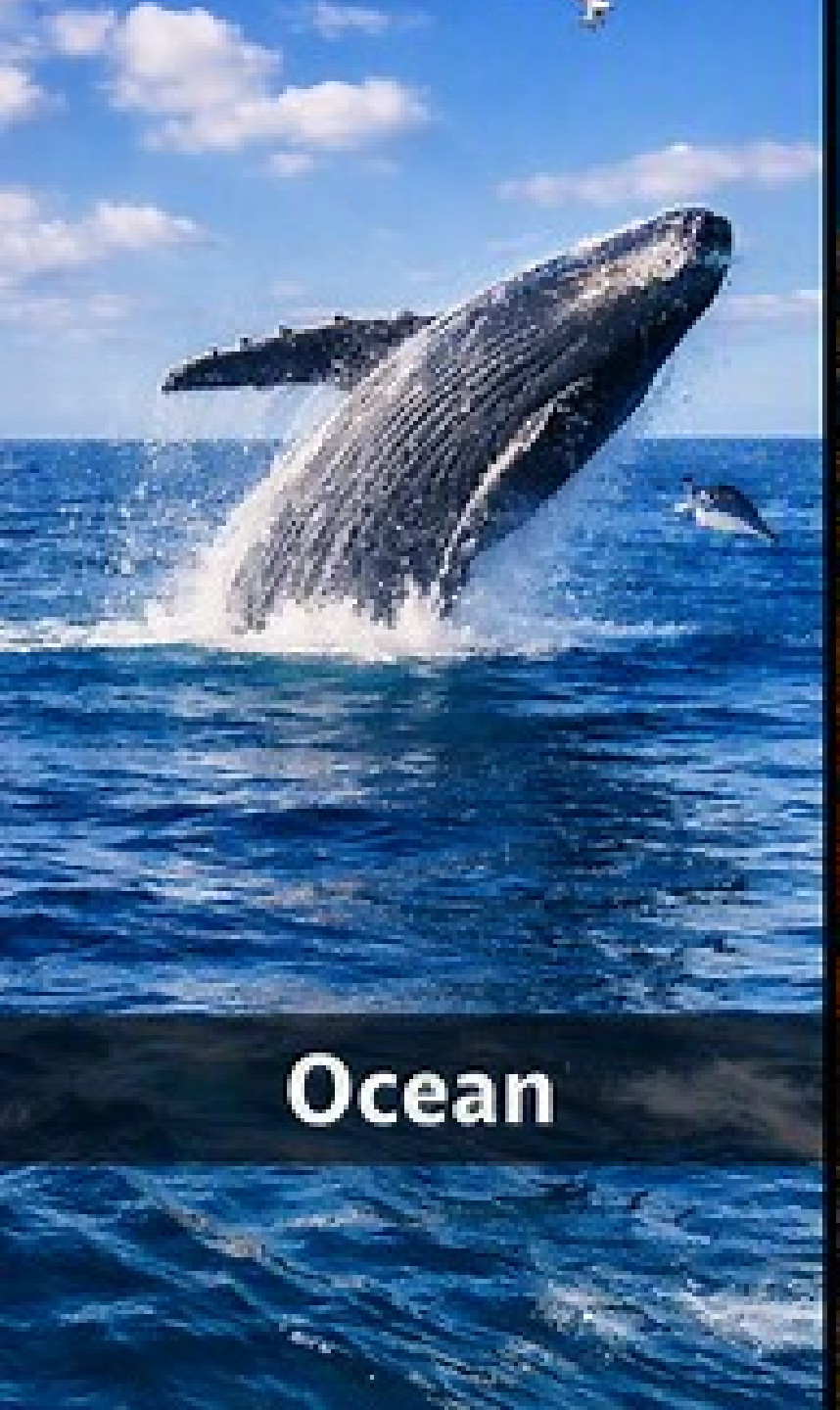
Wetlands (Agusan Marsh)

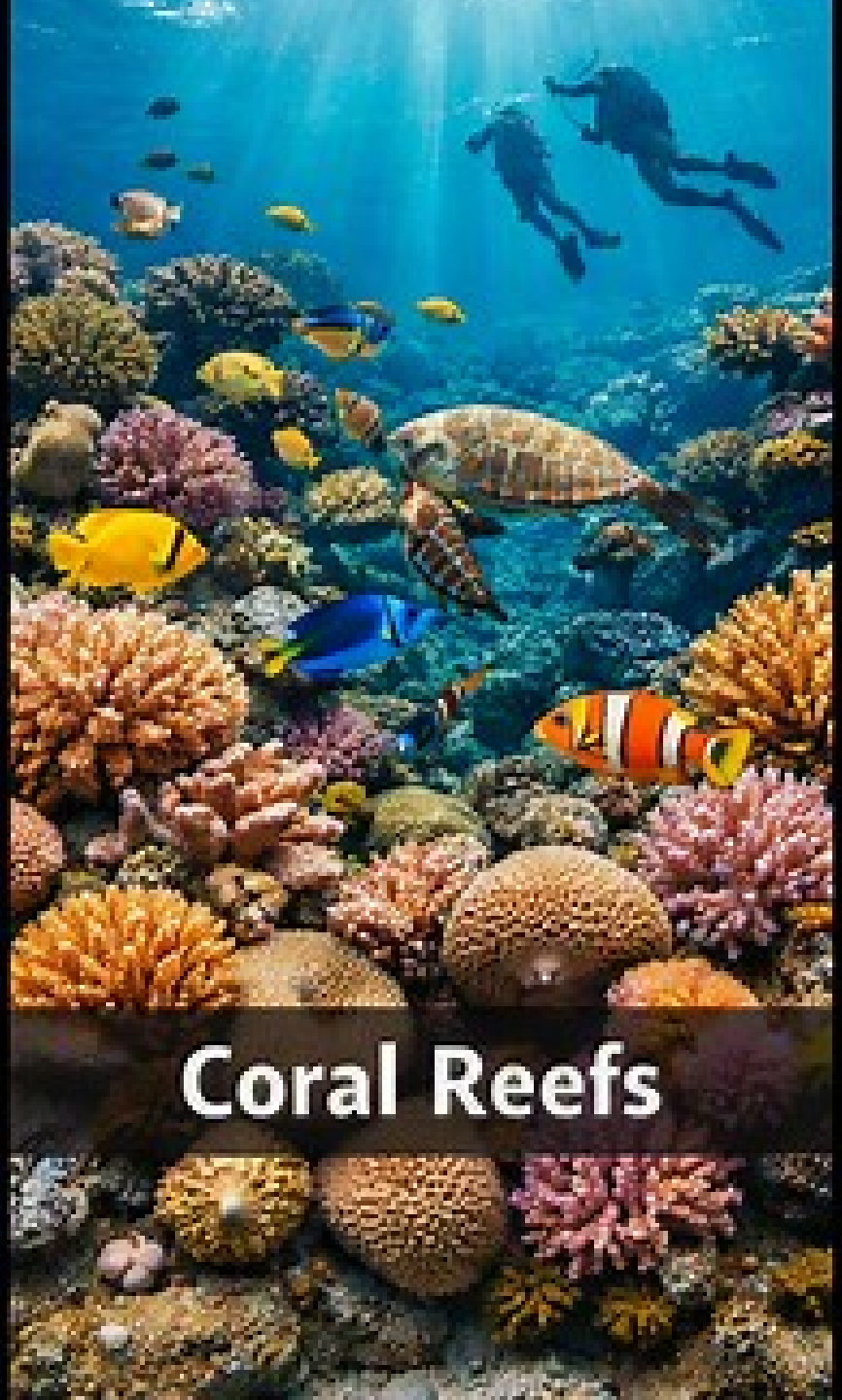
MARINE

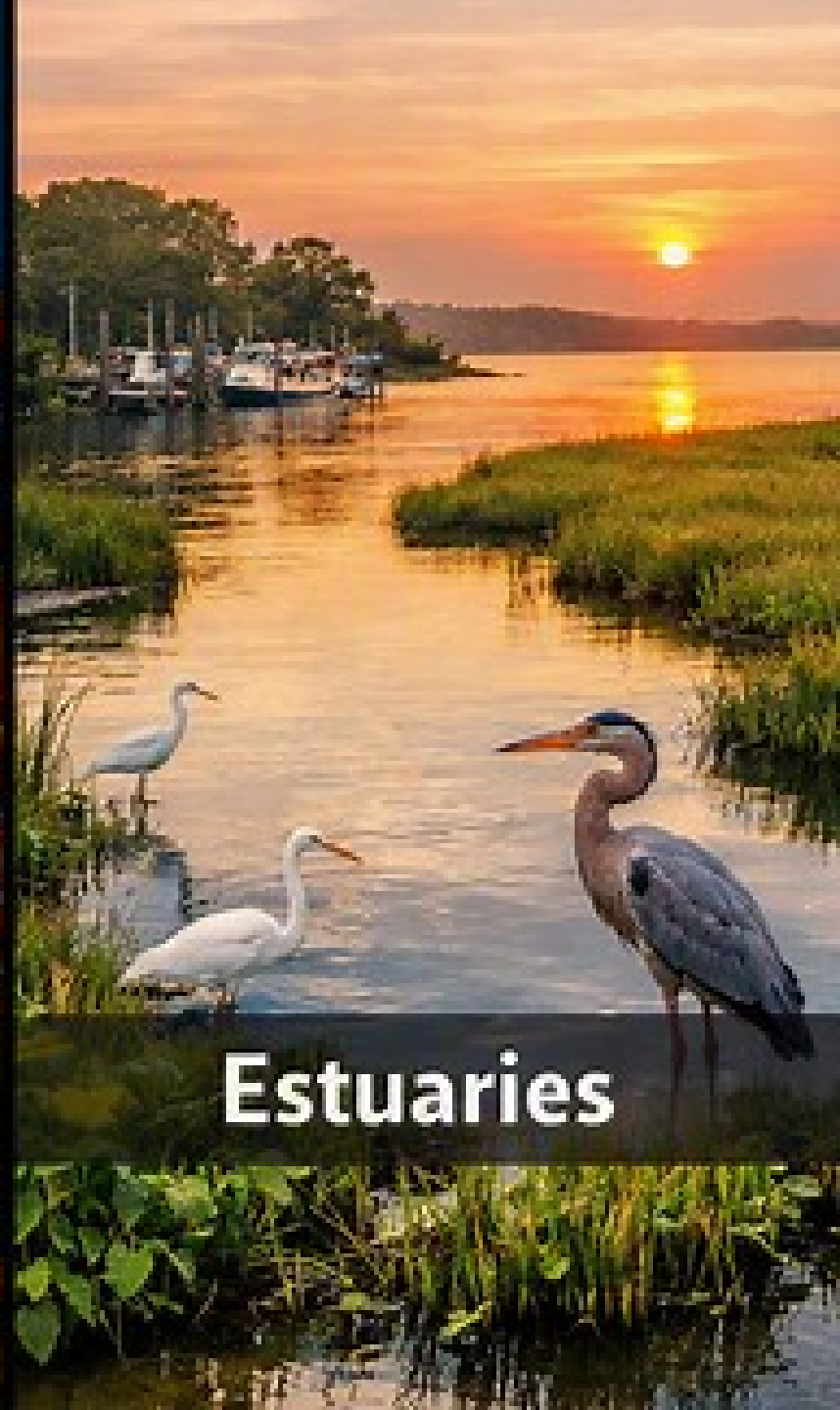
1. **Oceans:** Vast **saltwater** ecosystem covering most of Earth’s surface, **regulating climate** and supporting diverse life from plankton to whales.
2. **Coral Reefs: High biodiversity** ecosystems in **shallow, warm waters,** built by corals and **home** to countless fish and invertebrates.
3. **Estuaries: Transition zones** between **rivers and seas**, mixing freshwater and saltwater, serving as **nurseries** for many marine species.
    - Water is **brackish** — a mix of salt and fresh water.

MARINE: FLORA & FAUNA

Biome	Flora (Plants & Producers)	Fauna (Animals)
Ocean	Phytoplankton, seaweed, kelp forests, algae	Whales, dolphins, sharks, jellyfish, sea turtles, plankton
Coral Reefs	Zooxanthellae (symbiotic algae), sea grasses, algae	Clownfish, parrotfish, sea stars, reef sharks, sea urchins, corals (animals)
Estuaries	Mangroves, salt marsh grasses, algae	Crabs, oysters, shrimps, herons, egrets, fish fry (young fish)


Ocean


Coral Reefs


Estuaries

ENVIRONMENTAL FACTORS SHAPING BIOMES

1. **Climate**
   - Temperature and rainfall determine vegetation.
2. **Soil**
   - Nutrient availability affects plant growth.
3. **Sunlight**
   - Energy source for photosynthesis.
4. **Water**
   - Availability and salinity.

SUMMARY

- Biomes are shaped by **environmental factors**.
- Terrestrial biomes depend on **climate and soil**.
- Aquatic biomes depend on **water properties**.
- Understanding biomes helps us protect biodiversity.

TERRESTRIAL BIOMES

- **Tundra:** Cold, treeless, permafrost soil.
- **Taiga:** Coniferous forests, long winters.
- **Temperate Forest:** Four seasons, deciduous trees.
- **Grassland:** Dominated by grasses, few trees.
- **Desert:** Very dry, extreme temperatures.
- **Tropical Rainforest:** Warm, wet, high biodiversity.

AQUATIC BIOME

Freshwater

Low salinity, supporting diverse plants and animals adapted to non-salty environments.

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Marine

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DISCUSSION QUESTION

- How do human activities (deforestation, pollution, climate change) impact both terrestrial and aquatic biomes?

ASSIGNMENT

- **Title:** ***Biomes of the World – Digital Infographic***
- **Objective:** Students will design a digital infographic that explains one or more **world biomes.**

Content Requirements

- **Title & Subtitle** – Creative and clear.
- **Definition** – Concise description of the chosen biome.
- **Climate** – Temperature, precipitation, and seasonal patterns.
- **Flora & Fauna** – Representative plants and animals.
- **Human Impact** – Positive or negative effects. .
- **Visuals** – Icons, images, or diagrams to enhance understanding.
- **Sources** – Cite references for accuracy.

Format

- Digital layout (Canva, PowerPoint, or similar).
- Orientation: Landscape or Portrait.
- Size: A4 or Letter.
- Export format: PNG or PDF.

RUBRIC

Criteria	Excellent (10 pts)	Good (8 pts)	Fair (6 pts)	Needs Improvement (4 pts)
Content Accuracy	All biome info correct, complete, and relevant.	Minor inaccuracies or missing details.	Several inaccuracies or incomplete data.	Major errors or missing key concepts.
Visual Design	Creative, balanced, visually appealing layout.	Clear design with minor clutter.	Acceptable design but lacks harmony.	Poor layout or distracting visuals.
Organization & Clarity	Logical flow; easy to read and understand.	Mostly organized; minor confusion.	Some disorganization or unclear sections.	Hard to follow; lacks structure.
Creativity	Original, engaging, innovative presentation.	Some creative elements; moderately engaging.	Limited creativity; basic design.	Minimal effort; lacks originality.
Sources & Citations	All sources properly cited.	Some sources cited.	Few sources cited.	No citations provided.